Multi-Task Learning for Podcast Episode Understanding: Joint Emotion, Toxicity, and Category Analysis

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Abstract

Podcasts have emerged as a dominant audio medium with widespread reach and influence, yet they remain comparatively underexplored in computational research relative to text-based platforms. Fine-grained analysis of emotional tone, content safety, and topical characteristics at the episode level is critical for applications such as content recommendation, audience alignment, and brand-safe advertising. However, existing work largely focuses on ecosystem-level trends rather than detailed episode-level modeling. In this work, we propose a multi-level natural language processing pipeline for structured podcast content analysis that integrates emotion detection, toxicity estimation, and category classification. We apply `DistilRoBERTa` for emotion detection and `Toxic-BERT` for toxicity estimation to preprocessed podcast transcripts, adapting them to long-form, multi-speaker conversational data through custom preprocessing steps including audio artifact removal, consecutive-speaker merging, and turn segmentation. These models are used to generate pseudo-labels that enable large-scale supervision without manual annotation. Building on these signals, we develop a multi-task learning framework with a shared encoder (`DistilRoBERTa`) and task-specific heads that jointly optimizes episode category prediction, toxicity regression, and emotion modeling using a combined loss with learnable task weights. This shared representation allows the model to capture interactions between affective, safety-related, and topical dimensions of podcast content. We evaluate our approach on 907 podcast episodes comprising 123,379 turns from the Structured Podcast Research Corpus, comparing single-task and multi-task variants. Our results demonstrate meaningful correlations between emotional tone and toxicity as well as category-specific safety patterns. Overall, the findings validate multi-task learning as an effective and scalable approach for interpretable podcast episode analysis.

1 Introduction

With over 54% of American adults reporting podcast listening in the past 12 months—spanning diverse age groups from 67% of adults aged 18-29 to 33% of those 65 and older—podcasts have become a dominant medium for long-form audio content [Center, 2025]. Despite this massive reach and real-world impact on listeners' media consumption, purchasing decisions, and political engagement, podcasts remain significantly underexplored compared to text-based social media platforms like Twitter and Reddit. This gap exists largely due to technical challenges: podcasts are audio-only, distributed across multiple platforms with inconsistent metadata, and require automatic transcription to enable computational analysis.

Recent work has made progress in addressing this challenge. Litterer et al. [2024] introduce the Structured Podcast Research Corpus (SPoRC), a large-scale dataset of 1.1M podcast transcripts with speaker-level annotations and prosodic features, enabling ecosystem-level analysis of podcast content structure, community networks, and collective attention dynamics. However, SPoRC's focus remains on characterizing the podcast ecosystem as a whole rather than understanding individual episodes. Fine-grained analysis of emotional tone and content safety within episodes, which is important for podcast recommendation, audience alignment, and brand-safe advertising, remains limited.

In this work, we develop a multi-level NLP pipeline that combines emotion detection, toxicity classification, and category prediction to provide interpretable, turn-level and episode-level characterizations of podcast content. Our approach applies pretrained transformer models [Barbieri et al., 2021, AI, 2021] combined with custom preprocessing tailored to podcast-specific challenges, and develops a multi-task learning model that jointly predicts emotional distributions, safety labels, and episode categories from episode transcripts. This enables content platforms and advertisers to understand emotional dynamics and safety profiles of episodes at scale, while advancing the state of podcast-specific NLP research.

The rest of this paper is organized as follows: section 2 reviews related work on podcast analysis and multi-task learning. section 3 describes our data, preprocessing, and methodology. section 5 presents our experimental setup and results. section 6 provides in-depth analysis of our findings and discusses limitations of our approach. Finally, section 7 concludes with discussion of future work and potential improvements.

2 Related Work

The most directly relevant work is the Structured Podcast Research Corpus (SPoRC) introduced by [Litterer et al., 2024], which provides 1.1M podcast transcripts with speaker diarization, prosodic features, and inferred host/guest roles. Our work builds on SPoRC's infrastructure but shifts focus from ecosystem-level analysis (content structure, community networks, collective attention) to fine-grained speaker-turn analysis. Specifically, we apply emotion and toxicity detection at the turn level to characterize emotional dynamics and content safety within episodes.

Emotion recognition in conversational text has been addressed using transformer-based classifiers [Barbieri et al., 2021], and toxicity detection is well-studied in online platforms [Zhang et al., 2018, Pavlopoulos et al., 2020]. However, both tasks have primarily been applied to short social media posts rather than long-form multi-speaker dialogue. Our contribution is adapting these models to podcast transcripts through custom preprocessing and aggregating turn-level predictions to the episode level for interpretable content summaries.

3 Approach

Our analysis pipeline consists of three interconnected components operating at different granularity levels. First, we preprocess and aggregate the raw transcript data into clean episode-level documents. Next, we apply pretrained single-task models for emotion detection and toxicity classification at the turn level, aggregating results to the episode level. Finally, we develop a multi-task learning model that jointly predicts emotion distributions, toxicity scores, and podcast categories from episode transcripts, leveraging shared representations to improve performance across all tasks.

3.1 Data and Preprocessing

Our analysis uses two sample datasets from the Structured Podcast Research Corpus (SPoRC; [Litterer et al., 2024]): *episodeLevelDataSample.jsonl.gz* containing metadata for approximately 10,000 podcast episodes, and *speakerTurnDataSample.jsonl.gz* containing approximately 200,000 speaker turns from 1,033 unique episodes. We merge these datasets using the mp3url identifier and apply quality filters to obtain our analysis dataset.

3.1.1 Text Cleaning

We perform many basic text preprocessing steps to prepare transcripts for emotion and toxicity analysis. In short, this includes normalizing whitespace and lowercasing text, removing non-linguistic artifacts such as music markers (e.g., `[MUSIC]`), blank-audio indicators, ellipses, and transcription symbols, and eliminating punctuation. We apply English stopword removal using NLTK [**?**] to focus analysis on content-bearing terms.

To avoid noise from brief utterances (e.g., "yeah", "okay", "right") that remain after stopword removal, we retain only turns containing more than one word after filtering. This step eliminates approximately 15–20% of turns in a typical episode.

Some episodes contain single, very long speaker monologues that can exceed several thousand characters. Without segmentation, the emotion classifier would assign a single emotion label to an entire monologue despite potential shifts in tone and sentiment. To address this, we split turns exceeding 500 characters into contiguous 500-character chunks, each preserving speaker identity and episode metadata. This enables our emotion analysis to capture emotional progression within long contributions.

3.1.2 Data Aggregation

At the episode level, we collect attributes including title, podcast name, categories, and episode identifiers (mp3url) for dataset linking. Each episode in the dataset is annotated with up to ten hierarchical category labels (category1 through category10), where category1 represents the primary podcast category. For this work, we focus exclusively on predicting the primary category (category1), which contains 20 unique values. Across all ten category columns, the dataset contains 70 unique categories in total; however, we restrict our analysis to category1 to maintain a manageable classification target.

Transcripts frequently split uninterrupted speech from a single speaker across multiple sequential turns due to transcription segmentation heuristics. To maintain coherent speaker contributions, we merge consecutive turns from the same speaker by concatenating their text. This consolidation reduces turn count while creating more meaningful units for analysis.

After applying all preprocessing and filtering steps, our final dataset comprises 907 episodes that have speaker turn data in our sample, representing 123,379 total turns with an average of 135.5 turns per episode.

Table 1: Dataset Statistics

Metric	**Count**
Total Episodes	907
Total Turns	123,379
Average Turns per Episode	135.5
Unique Categories	20
Emotion Classes	7

3.2 Emotion Detection

We use a pretrained DistilRoBERTa model fine-tuned for emotion classification [Barbieri et al., 2021] (model: `j-hartmann/emotion-english-distilroberta-base`). This model classifies text into seven emotion categories: anger, disgust, fear, joy, neutral, sadness, and surprise. We apply the model at the speaker-turn level, feeding each turn's preprocessed text and recording the predicted emotion label and confidence score. Episode-level emotion distributions are computed by aggregating predictions across all turns in an episode, enabling comparison of emotional tone across podcast categories and genres.

3.2.1 Emotion Distribution Results

Across the 123,379 turns in our dataset, the emotion distribution is heavily skewed toward neutral expressions. Specifically, neutral emotions dominate at 44.24%, followed by joy (18.87%) and surprise (15.75%), which together comprise 78.9% of all turns. Conversely, negative emotions are less prevalent: sadness accounts for 8.96% of turns, anger for 6.64%, fear for 4.36%, and disgust for the smallest proportion at 1.17%. This distribution aligns with expectations for conversational podcast content, where most exchanges maintain a neutral tone interspersed with moments of humor or positive affect. Graph of detailed emotion proportions are shown in Appendix Figure 4.

3.2.2 Category-Emotion Coupling

Emotional expressions vary systematically across podcast categories, revealing meaningful category-emotion couplings. News podcasts exhibit the highest anger probability (15.3%) followed by true crime (10.0%) and society (8.7%), reflecting journalism's focus on contentious issues, while leisure (44.7%), science (44.5%), and music (42.0%) maintain the strongest neutral tone. Education and religion rank highest in joy (26.6% and 25.1% respectively), suggesting informational content with positive framing. Technology podcasts show the highest surprise probability (24.9%), reflecting rapid innovation and novel developments. These patterns are visualized in Appendix Figure 9 and raw results can be seen in Figure 9.

3.3 Brand Safety Analysis

We employ the `unitary/toxic-bert` transformer-based binary toxicity classifier [AI, 2021] to identify potentially unsafe or offensive content. The model is applied to individual speaker turns with a confidence threshold of 0.5 and turns scoring above this are classified as toxic. At the episode level, we compute: (1) the proportion of toxic turns, (2) the maximum toxicity score, and (3) the mean toxicity score. If an episode's maximum toxicity score exceeds 0.15, that episode is flagged as unsafe. This provides content moderators with both fine-grained (turn-level) and summary (episode-level) views for brand safety assessment.

3.4 Single-Task Classification Model

As a baseline, we implement a single-task episode-level classification model that predicts the primary category of a podcast episode using only its transcript text. The model consists of a pre-trained `distilroberta-base` encoder followed by a lightweight task-specific classification head. Given an episode transcript, the encoder produces contextualized token representations, from which the hidden state corresponding to the special `[CLS]` token is used as a fixed-length summary of the episode.

This pooled representation is passed through a dropout layer for regularization and then fed into a linear layer that outputs logits over the category classes. The model is trained end-to-end using a cross-entropy loss on the category labels. This single-task model serves as our baseline, allowing us to evaluate whether multi-task learning improves performance.

3.5 Multi-Task Learning Model

To take advantage of the connections between emotion, toxicity, and category, we build a single model that predicts all three. This is more efficient than three separate models and lets them learn from each other.

3.5.1 Architecture

The model uses `DistilRoBERTa` as a shared encoder, which processes the concatenated episode transcript and produces contextual embeddings. Three task-specific heads are attached to this shared representation, each implemented as a two-layer sequential network.

First we have a category head which has of a two-layer network consisting of a fully connected layer mapping the encoder hidden size H to H, followed by ReLU activation, dropout, and a final linear layer outputting logits for K podcast categories.

Then we have a toxicity head, also a two-layer network with identical structure (linear, ReLU, dropout, linear) that outputs a single continuous score in the range $[0, 1]$ representing episode-level toxicity. Instead of just classifying content as safe or unsafe, this head outputs a toxicity score that shows how much harmful content is in each episode.

Finally we have the emotion head, also a two-layer network that outputs a 7-dimensional softmax distribution over emotions (anger, disgust, fear, joy, neutral, sadness, surprise), allowing the model to capture the full emotional landscape of each episode rather than a single dominant emotion.

3.5.2 Label Generation

Since manual annotations are unavailable, we generate pseudo-labels from our pretrained single-task models. Category labels are directly extracted from episode-level metadata using the `category1` field. Toxicity labels are aggregated from cached turn-level toxicity scores and we compute the maximum toxicity score across all turns in each episode to create a continuous label representing the highest risk level. Emotion labels are soft labels computed as the normalized distribution of emotion frequencies across all turns in the episode.

3.5.3 Joint Training

The multi-task model is trained end-to-end using a joint objective that balances the contributions of all three tasks. The total training loss is defined as

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cat}}\, \mathcal{L}_{\text{CE}}^{\text{cat}} + \lambda_{\text{tox}}\, \mathcal{L}_{\text{MSE}}^{\text{tox}} + \lambda_{\text{emo}}\, \mathcal{L}_{\text{KL}}^{\text{emo}}, \tag{1}$$

where $\mathcal{L}_{\text{CE}}^{\text{cat}}$ denotes the cross-entropy loss for episode category classification, $\mathcal{L}_{\text{MSE}}^{\text{tox}}$ is the mean squared error loss used for episode-level toxicity regression, and $\mathcal{L}_{\text{KL}}^{\text{emo}}$ is the KL divergence between the predicted and target emotion distributions. The coefficients λ_{cat}, λ_{tox}, and λ_{emo} control the relative contribution of each task.

In addition to fixed loss weighting, we experiment with learnable uncertainty-based weighting (Kendall et al. [2017]), in which task weights are treated as trainable parameters rather than manually selected hyperparameters. Following prior work on multi-task learning, each task is associated with a learned uncertainty parameter that dynamically rescales its corresponding loss term during training. This approach allows the model to automatically balance tasks with different noise levels and learning dynamics, reducing sensitivity to manual hyperparameter tuning.

By jointly optimizing these objectives, the shared encoder learns representations that capture both explicit task-specific signals (category and toxicity) and implicit relationships between emotional content, safety characteristics, and category structure. This allows the model to learn connections between tasks rather than treating them independently.

4 Exploratory Analysis of Pre-trained Models

4.0.1 Emotion Distribution Results

Across the 123,379 turns in our dataset, the emotion distribution is heavily skewed toward neutral expressions. Specifically, neutral emotions dominate at 44.24%, followed by joy (18.87%) and surprise (15.75%), which together comprise 78.9% of all turns. Conversely, negative emotions are less prevalent: sadness accounts for 8.96% of turns, anger for 6.64%, fear for 4.36%, and disgust for the smallest proportion at 1.17%. This distribution aligns with expectations for conversational podcast content, where most exchanges maintain a neutral tone interspersed with moments of humor or positive affect. Graph of detailed emotion proportions are provided in Appendix Figure 4.

4.0.2 Toxicity Distribution Results

Across the dataset, toxicity is not uniformly distributed. The median toxicity ratio is 0.004 with a mean of 0.037, indicating that toxicity is concentrated in a minority of high-risk episodes rather than spread broadly across the corpus. Overall, 93.9% of episodes are classified as safe (maximum toxicity score $\leq$ 0.15), demonstrating that the majority of podcast content is brand-appropriate. The distributions of toxicity ratio, mean toxicity score, and maximum toxicity score are shown in Appendix Figure 1, Figure 2, and Figure 3.

4.0.3 Category and Emotion Coupling

Emotional expressions vary systematically across podcast categories, revealing meaningful category-emotion couplings. News podcasts exhibit the highest anger probability (15.3%) followed by true crime (10.0%) and society (8.7%), reflecting journalism's focus on contentious issues, while leisure (44.7%), science (44.5%), and music (42.0%) maintain the strongest neutral tone. Education and religion rank highest in joy (26.6% and 25.1% respectively), suggesting informational content with positive framing. Technology podcasts show the highest surprise probability (24.9%), reflecting rapid innovation and novel developments. These patterns are visualized in Appendix Figure 9 and raw results can be seen in Figure 9.

4.0.4 Toxicity and Emotion Correlations

Toxicity detection reveals important associations between emotional tone and unsafe content. Across all emotions, disgust shows the highest mean toxicity score (0.369), followed closely by anger (0.359), indicating that negative emotional expressions correlate with potentially offensive language. By contrast, neutral (0.035), joy (0.049), and surprise (0.071) expressions exhibit substantially lower mean toxicity scores, suggesting that positive or neutral conversational exchanges are mainly brand-safe. Detailed toxicity-emotion relationships are shown in Appendix Figure 5.

4.0.5 Category and Toxicity Correlations

At the category level, significant variation emerges in content safety. News podcasts exhibit the highest average toxicity ratio (13.5%), followed by Comedy (12.5%), reflecting the often contentious nature of political commentary and satirical content. Educational and technology-focused content shows minimal toxicity (approximately 1%), while health and government categories average under 1% toxicity ratio. This category-level breakdown is visualized in Appendix Figure 6. Furthermore, the severity of toxic content (maximum toxicity scores) varies by category, with News and Comedy also showing the highest maximum toxicity exposure; detailed maximum toxicity by category is provided in Appendix Figure 7. Safety percentage by category (Appendix Figure 8) confirms these findings, showing that educational, technology, and health categories maintain near-universal safety ratings above 99%.

5 Experiments

5.1 Dataset and Preprocessing

We test our models on podcast episodes, each represented as a sequence of speaker turns and annotated with a primary category label (`category1`). In addition to the category labels, we use pre-trained models to automatically generate emotion and toxicity labels, so we don't need to annotate everything manually.

We only include categories with at least two episodes and split the data 80/20 for training and validation. Episodes vary a lot in length—some have over 1,000 turns. Since we can't fit entire episodes in memory, we randomly sample a fixed window of turns during training and limit each turn to 128 characters.

5.2 Models and Training Setup

All models share the same backbone encoder, `distilroberta-base`. We compare a single-task baseline against a multi-task model that shares the encoder across tasks and uses separate task-specific heads. The single-task model predicts the primary category only. The multi-task model jointly predicts (1) the primary category (multi-class classification), (2) episode-level toxicity (regression), and (3) episode-level emotion distribution (soft labels based on aggregated turn-level emotions).

Unless otherwise specified, all models are trained for 10 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 4. Due to memory constraints associated with long transcripts, training is performed using a windowed representation in which each episode is limited to 32 turns per window, with each turn truncated to a maximum of 128 characters. During training, we represent each episode using fixed-size windows of 32 turns. During evaluation and inference, we process each episode in overlapping windows of 32 turns and average predictions across windows to obtain episode-level metrics.

5.3 Evaluation Metrics

We use different metrics for each task. For category prediction, we report accuracy and macro-F1 score. We care more about macro-F1 because some categories are much more common than others, so we don't want the model just memorizing the frequent ones. Toxicity is evaluated as a regression task using mean absolute error (MAE). We measure emotion performance with macro-F1, but we don't weight it as heavily as category prediction. It's more of a check that the model is learning emotions, not our primary focus.

For convenience in selecting checkpoints and comparing multi-task variants, we also report a composite score defined as

$$\text{OverallScore} = \alpha \cdot \text{F1}_{\text{cat}} + \beta \cdot (1 - \text{MAE}_{\text{tox}}) + \gamma \cdot \text{F1}_{\text{emo}}, \tag{2}$$

where $\alpha, \beta, \gamma \geq 0$ control the relative contribution of each task. In our experiments, we set $\alpha = \beta = \gamma = \frac{1}{3}$ to give equal weight to all three tasks during model selection. We note that this composite is used as a model-selection heuristic and does not replace reporting task-specific metrics.

We use a composite score mainly to pick the best model during training and compare different versions. It's not a real performance metric though. The tasks are too different to combine them meaningfully.

5.4 Experimental Results

5.4.1 Experiment 1: Single-Task vs. Multi-Task Learning

This experiment tests whether auxiliary supervision (toxicity and emotion) improves category classification. The single-task baseline is trained to predict episode category only, while the multi-task model jointly optimizes category, toxicity, and emotion objectives. Both models use the same encoder architecture, split, optimizer, and training budget The only difference is the inclusion of auxiliary heads and losses in the multi-task setting.

Table 2 reports validation performance. Notably, the multi-task model achieves higher macro-F1 despite lower accuracy, indicating improved performance on minority categories at the expense of some majority-class predictions.

Table 2: Validation performance for single-task category classification vs. multi-task learning.

Model	Accuracy ↑	Macro-F1 ↑
Single-task (category only)	0.780	0.561
Multi-task (3 heads)	0.690	0.598

5.4.2 Experiment 2: Fixed vs. Learnable Loss Weighting

Multi-task training requires balancing losses across tasks. We compare a fixed-weight strategy (manually chosen task weights) against a learnable weight strategy in which task weights are optimized jointly with model parameters. All other training settings are held constant. Table 3 summarizes the averaged validation results across runs.

Table 3: Comparison of fixed vs. learnable loss weighting for the multi-task model (validation).

Weighting	Cat Acc ↑	Cat Macro-F1 ↑	Tox MAE ↓	Emo Macro-F1 ↑	OverallScore ↑
Learnable weights	0.690	0.598	0.189	0.344	0.5839
Fixed weights	0.685	0.597	0.201	0.414	0.6033

6 Analysis

6.1 Single-Task vs. Multi-Task Trade-offs

Experiment 5.4.1 reveals an important trade-off: the multi-task model improves macro-F1 for category prediction while reducing overall accuracy. This pattern is consistent with a shift toward better performance on under-represented classes. Accuracy is dominated by frequent categories, while macro-F1 treats all categories equally. This means the model can improve on rare categories and still see its overall accuracy drop. This suggests that auxiliary supervision from toxicity and emotion encourages the shared encoder to learn representations that generalize beyond the dominant category patterns.

A practical implication is that model choice should depend on the target application. If the system prioritizes overall correctness on frequent categories, the single-task model may be preferable. If the objective is robust performance across all categories, including minority labels, the multi-task model is more suitable.

6.2 Effect of Loss Weighting

Experiment 5.4.2 indicates that the choice of loss weighting influences both optimization stability and task trade-offs. The learnable-weight approach improves toxicity MAE, suggesting that the optimization procedure assigns greater emphasis to toxicity regression when it benefits the overall training objective. However, fixed weights yield higher emotion macro-F1 and a higher composite OverallScore. This disparity highlights that learnable weighting can implicitly favor tasks with larger gradients or easier optimization dynamics, which may not align with the desired downstream priorities. In future work, the weighting mechanism could be constrained (e.g., bounded weights or task-normalized losses) to avoid over-emphasizing a single auxiliary objective.

6.3 Error Characteristics and Limitations

Several factors may limit model performance. First, category labels may be noisy or overlapping, particularly for episodes that cover multiple themes, which introduces ambiguity into the classification task. Second, emotion and toxicity supervision are derived from external models and cached predictions rather than human annotations, and may therefore propagate systematic biases or errors into the training process. Third, our token limit means we truncate long turns, which could cause us to lose important context, especially in longer conversational turns.

In addition, although we explored larger turn-window sizes to provide broader contextual coverage, practical memory constraints prevented reliable training with windows larger than 32 turns, as these settings consistently resulted in out-of-memory errors on the available hardware. Consequently, the model is trained using relatively small sampled windows, which may omit highly localized but informative cues. This limitation motivates the use of multi-window aggregation at inference time to better approximate full-episode behavior.

Overall, the experiments suggest that multi-task learning is a viable strategy for improving balanced category performance, but its effectiveness depends on careful loss balancing and windowing choices under the constraints imposed by long-form sequence modeling.

7 Conclusion

In this project, we investigated the problem of podcast episode understanding through episode-level category classification, augmented with auxiliary supervision derived from emotional and toxicity signals. In addition to training predictive models, we first constructed pseudo-labels for emotion and toxicity by applying pre-trained detectors to episode transcripts. These automatically generated labels enabled large-scale supervision without the need for manual annotation and formed the basis for subsequent multi-task learning experiments.

Building on these pseudo-labels, we proposed and evaluated a multi-task learning framework that jointly optimizes episode category classification, toxicity regression, and emotion prediction using a shared transformer encoder. Experimental results show that multi-task learning improves balanced classification performance compared to a single-task baseline, as reflected by higher macro-averaged F1 scores. While overall accuracy may decrease slightly due to trade-offs with majority classes, the multi-task model demonstrates stronger generalization across under-represented categories.

Our experiments further highlight the importance of architectural and training design choices when modeling long-form conversational data. In particular, memory constraints required training on sampled windows of turns rather than full episodes, motivating the use of window-based aggregation during evaluation. We also observed that loss weighting strategies influence task trade-offs, with learnable weights providing stable optimization but only small gains over carefully tuned fixed weights.

Several limitations remain. Emotion and toxicity supervision are derived from external models and may introduce noise or bias into the training process. Additionally, the reliance on fixed-size windows may omit highly localized cues, especially in long episodes. Future work could explore long-context architectures, such as sparse-attention or hierarchical transformers, to better capture global episode structure. Incorporating human-annotated emotion and toxicity labels, as well as adaptive window selection strategies, could further improve robustness and interpretability.

Overall, this work demonstrates that combining pseudo-label generation with multi-task learning is a practical and effective approach for structured understanding of long-form podcast content, and provides a foundation for more advanced models operating on extended conversational data.

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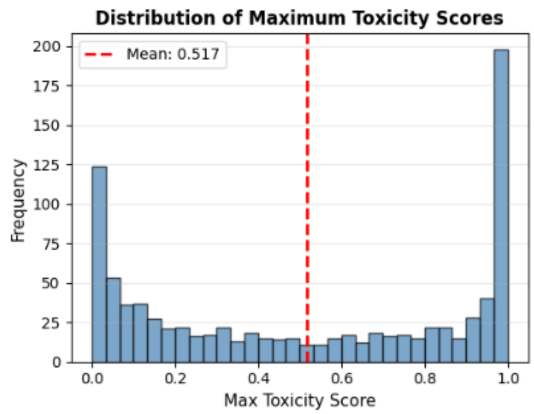

Figure 1: Distribution of Maximum Toxicity Scores

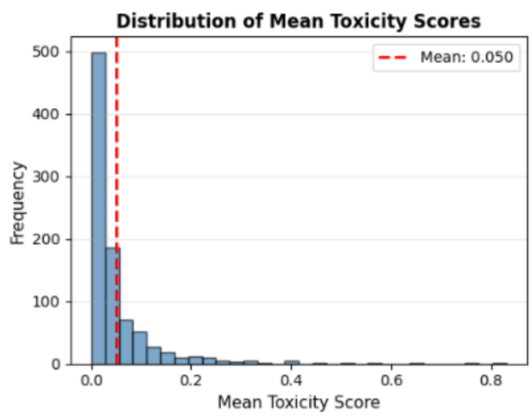

Figure 2: Distribution of Mean Toxicity Scores

A Appendix

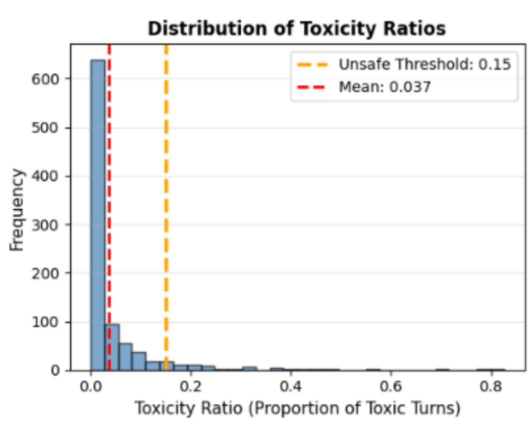

Figure 3: Distribution of Toxicity Ratio

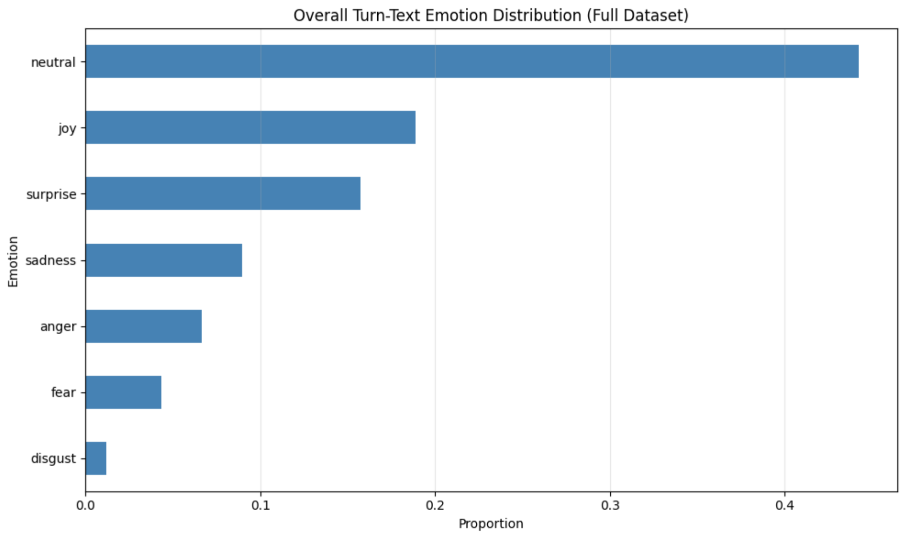

Figure 4: Overall Turn-Text Emotion Distribution

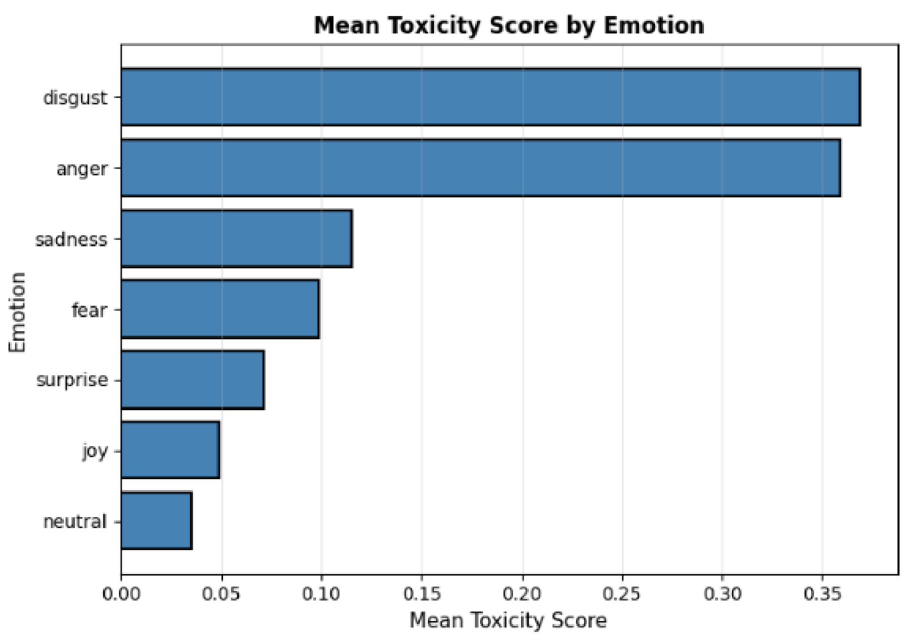

Figure 5: Mean Toxicity Score by Emotion

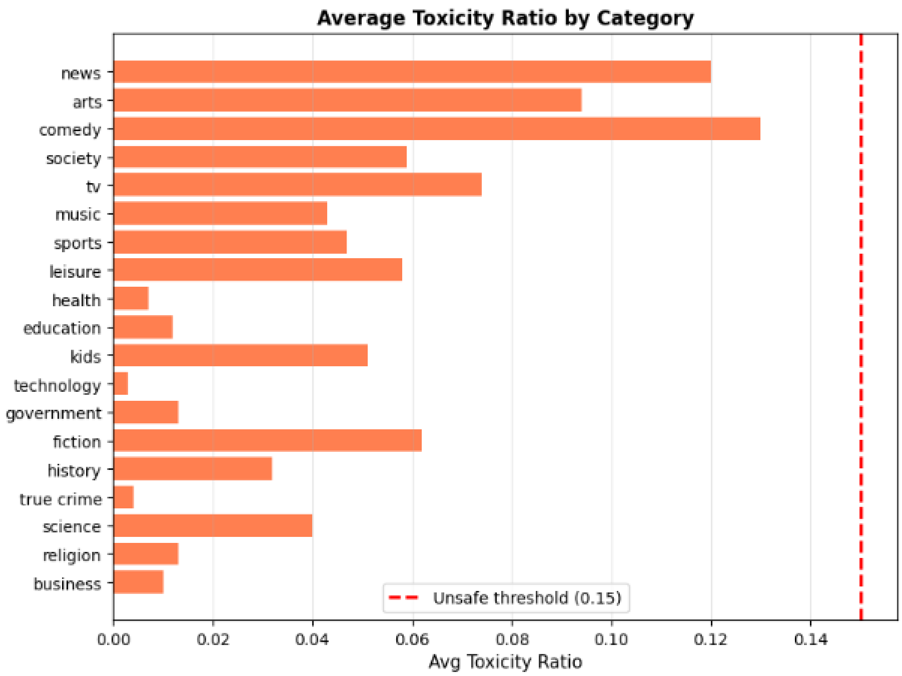

Figure 6: Average Toxicity Ratio by Category

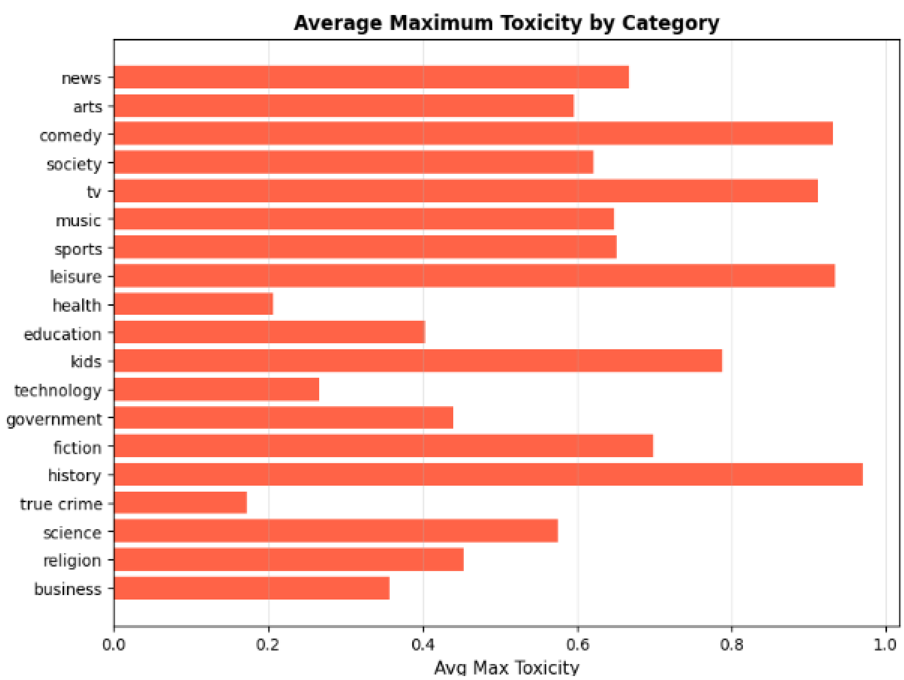

Figure 7: Average Maximum Toxicity by Category

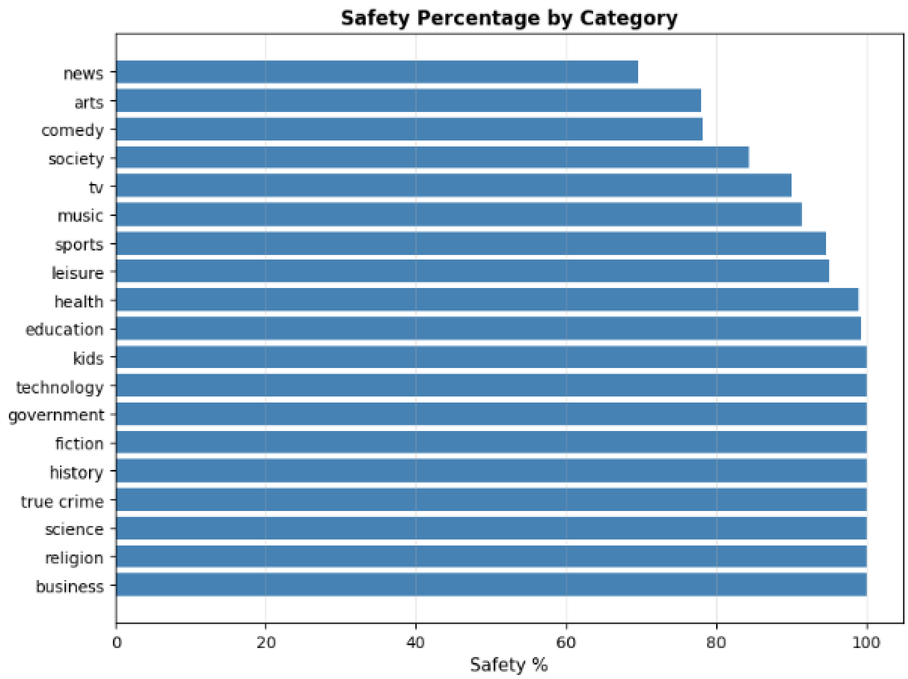

Figure 8: Safety Percentage by Category

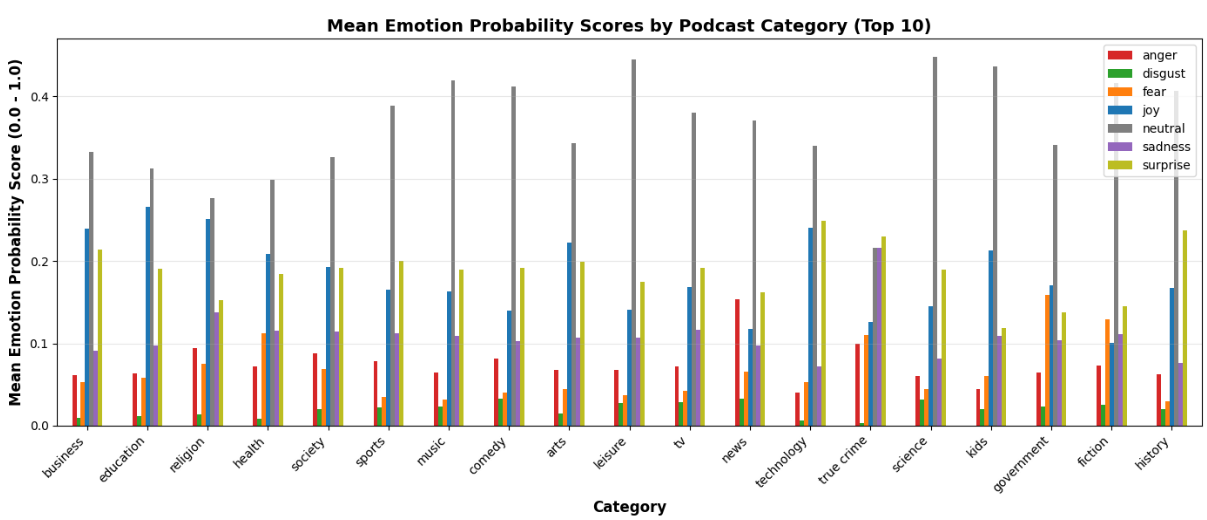

Figure 9: Emotion Distribution by Category

Category	Probabilty	Episodes
News	15.3%	23
True Crime	10.0%	5
Religion	9.4%	114
Society	8.7%	96
Comedy	8.1%	46

(a) Anger

Category	Probabilty	Episodes
Comedy	3.3%	46
News	3.3%	23
Science	3.2%	4
TV	2.9%	30
Leisure	2.8%	41

(b) Disgust

Category	Probabilty	Episodes
Government	15.9%	3
Fiction	12.9%	3
Health	11.3%	97
True Crime	11.0%	5
Religion	7.5%	114

(c) Fear

Category	Probabilty	Episodes
Education	26.6%	130
Religion	25.1%	114
Technology	24.0%	8
Business	23.9%	157
Arts	22.3%	41

(d) Joy

Category	Probabilty	Episodes
Science	44.7%	4
Leisure	44.5%	41
Kids	43.6%	4
Music	42.0%	47
Fiction	41.6%	3

(e) Neutral

Category	Probabilty	Episodes
True Crime	21.6%	5
Religion	13.8%	114
TV	11.6%	30
Health	11.5%	97
Society	11.4%	96

(f) Sadness

Category	Probabilty	Episodes
Technology	24.9%	8
History	23.7%	1
True Crime	22.9%	5
Business	21.4%	157
Sports	20.1%	57

(g) Surprise

Figure 10: Top 5 Categories by Emotion Probability